MAgent Theory 2.0

Link Dynamics:
Dictionaries in the Two Free Energies

Robert C. Dennis

Independent Researcher

October 2026

Status and summary

Status. This working document is superseded by the main manuscript's chapter on learning directed links and its companion chapter, which are authoritative for every statement and for every label that this document shares with them. The chapter also contains the general-law row objective with explicit row entropy, the coupled Gaussian flow of all means, covariances and rows, its global existence and convergence theorem, the three-agent threshold and the local pitchfork, none of which is reproduced here. This document now carries the chapter's corrections of the per-agent split, the Gibbs-row derivative, the source-role stationarity condition, the scalar certainty statement and the open problems. Later refinements of the chapter are not repeated here; they include the group-transitivity form and compact counterpart of the pointwise interpretation gauge, the uniqueness of the law-level covariance fixed point, the per-agent instability example, the measurability of the receiver kernel, the statement that this per-agent model is a record-likelihood model distinct from the mixture-source model, and the attributions to radical interpretation, Gaussian belief propagation and persuasion bias. This document preserves the standalone derivation of receiver-owned link dynamics. The calculations distinguish a normalized global model, normalized per-agent models with peers held as data, and a reciprocal potential on laws.

The dictionary derivatives agree with the global derivatives only in the plug-in per-agent model. Integrating the peers' laws instead widens both record noise and loop factors. Equality of objective derivatives does not imply equality of Fisher velocities unless the recognition metrics also agree. The linear mean-flow results hold with the stated Gaussian covariances and source rows fixed. The grounded-limit theorem covers the global and plug-in models; it does not cover coupled law-level covariance updates.

Comparison records confound changes in content with changes in the dictionary. Gaussian conditioning resolves a discrepancy among receiver adjustment, sender adjustment, and reinterpretation according to their prior covariances. A directed ring gives a non-gradient per-agent flow; its weak-anchor coupling vanishes only at nonzero operating content. A single loop gives an exact example of overconfidence from factorization and recycled peer information. The reciprocal potential has a source-role covariance derivative, but increasing stationary variance with disagreement is proved only in the stated scalar regime. General-law extensions remain open; the joint Gaussian mean, covariance and row dynamics are treated in the main manuscript chapter.

Contents

Status and summary i

1 The tangent communication model 1

2 What conversation determines 3

3 The two flows 5

4 The grounded limit 8

5 Confidence under ownership 9

6 The reciprocal potential 11

7 Scope and open problems 14

Chapter 1

The tangent communication model

Two agents who exchange messages about something they both represent can register whether a message matches what the receiver expected. A comparison record is a noisy linear function $A_i y_i - B_{ij} y_j + \epsilon$ of the two agents' contents. With codes and receiver-owned dictionaries the natural choice is $B_{ij} = \bar{T}_e$, the sender's code, and $A_i = \bar{T}_e \bar{\theta}_e^{-1}$, the receiver's prediction of the message from its own content translated into the sender's coordinates. The following construction linearizes that record and places it in the two free energies.

Definition 1.1 (Tangent communication model). Let $\mathcal{V}$ be a finite set of agents and $\mathcal{E}$ a set of directed edges $e : j \to i$ with receiver i and sender j, and write $\mathrm{in}(i)$ for the edges received by i. Agent i has content $y_i \in \mathbb{R}^K$ in its reference coordinates. Edge e carries the receiver's dictionary $\bar{\theta}_e = e^{\omega_e}$ with $\omega_e \in \mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$, the sender's attending code $\bar{T}_e : \mathbb{R}^K \to H_e = \mathbb{R}^{r_e}$ of full row rank $r_e \le K$ and a positive-definite noise covariance R_e. Fix operating contents $c_i \in \mathbb{R}^K$ and write $\kappa_i \omega = \omega c_i$. The dictionary is a receiver-owned hypothesis about the sender's coordinates; the independently owned opposite dictionary need not be its inverse. Define $N_e = \{\delta \in \mathfrak{g} : \bar{T}_e \delta = 0\}$, the directions invisible to grounded referents that span the content space. These can be fewer than the directions invisible at a single operating content, which satisfy only $\bar{T}_e \delta c_i = 0$. The comparison record of e is

$$o_e = \bar{T}_e\big(y_i - y_j - \kappa_i \omega_e\big) + \varepsilon_e, \qquad \varepsilon_e \sim \mathcal{N}(0, R_e), \tag{1.1}$$

the first-order form of $\bar{T}_e \bar{\theta}_e^{-1} y_i - \bar{T}_e y_j + \varepsilon_e$ at $\bar{\theta}_e = I$ and $y_i = c_i$. Each agent has a self term $\mathcal{N}(m_i, P_i^{-1})$ with P_i positive definite, the Gaussian form at the operating point of its own generative model and its own records. Each edge has a grounded term $\exp(-\frac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e})$ with G_e positive semidefinite, the information of grounded messages, with $\ker G_e = N_e$ when their referents identify every coded direction. The dictionaries have an anchored loop prior with anchor $\sigma^{-2} I$ on each edge, a finite set $\mathcal{L}$ of simple directed loops containing every two-cycle, linearized holonomy $(B\omega)_\gamma = \sum_{e \in \gamma} \omega_e$, positive-definite forms W_γ and weight $\lambda \ge 0$. Put $W = \bigoplus_\gamma W_\gamma$ and use $\|v\|^2_A = v^T A v$. All codes, operating contents, grounding data, and prior forms are fixed. The Frobenius anchor and loop forms are declared reference-coordinate data, not an assertion of a positive-definite conjugation-invariant form on $\mathfrak{gl}(K, \mathbb{R})$.

Definition 1.2 (Global joint). Write $x = (y, \omega)$ and $L_e x = \bar{T}_e(y_i - y_j - \kappa_i \omega_e)$. The global joint has posterior density proportional to $e^{-E(x)}$ with

$$\begin{aligned} E(x) = &\sum_i \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_e \Big(\tfrac{1}{2\sigma^2}\|\omega_e\|^2 + \tfrac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e} + \tfrac{1}{2}\|o_e - L_e x\|^2_{R_e^{-1}} \Big) \\ &+ \tfrac{\lambda}{2} \sum_{\gamma \in \mathcal{L}} \|(B\omega)_\gamma\|^2_{W_\gamma}. \end{aligned} \tag{1.2}$$

Its precision J is positive definite because the self terms and the anchor are, so the joint is normalized for every value of the registered records. Write η for its information vector and $x^\star = J^{-1}\eta$ for the posterior mean.

The linearization drops the product $\omega_e(y_i - c_i)$, which is of second order in the deviations from the operating point. Registering $o_e = 0$ registers agreement. The record is invariant under passive changes of frame when the displayed contents, codes, and dictionaries are transformed together, because the dictionary sits in the receiver's matrix.

Definition 1.3 (Per-agent tangent model). Fix agent i and the current peer laws, $q_k = \mathcal{N}(\mu_k, \Sigma_k)$ for contents and $r_f = \mathcal{N}(a_f, V_f)$ for the dictionaries that i does not own. Agent i's model has the variables $x_i = (y_i, \omega_{\mathrm{in}(i)})$ and negative log density, up to a constant,

$$\begin{aligned} E_i = {} & \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_{e\in\mathrm{in}(i)} \Big(\tfrac{1}{2\sigma^2}\|\omega_e\|^2 + \tfrac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e} + \tfrac{1}{2}\big\|o_e - \bar{T}_e(y_i - \mu_j - \kappa_i\omega_e)\big\|^2_{\tilde{R}_e^{-1}} \Big) \\ & + \sum_{\gamma\in\mathcal{L}_i} \tfrac{1}{2}\Big\| \sum_{e\in\gamma\cap\mathrm{in}(i)} \omega_e + \sum_{f\in\gamma\setminus\mathrm{in}(i)} a_f \Big\|^2_{\Omega_{i\gamma}}, \end{aligned} \tag{1.3}$$

where $\mathcal{L}_i$ is the set of loops through an edge of i and

$$\tilde{R}_e = R_e + \bar{T}_e \Sigma_j \bar{T}_e^T, \qquad \Omega_{i\gamma} = \Big((\lambda W_\gamma)^{-1} + \sum_{f\in\gamma\setminus\mathrm{in}(i)} V_f \Big)^{-1}. \tag{1.4}$$

For $\lambda > 0$, these are the record eq. (1.1) and the loop factor integrated against the peers' laws; we call this the law-level model. At $\lambda = 0$ the loop terms are omitted, equivalently $\Omega_{i\gamma} = 0$ by continuity. The convolution uses the declared product of peer dictionary marginals for each loop factor. Replacing $\tilde{R}_e$ by R_e and $\Omega_{i\gamma}$ by λW_γ gives the plug-in model, in which peers enter only through their means.

The integration is Gaussian convolution: $\int \mathcal{N}(o; \bar{T}(y_i - y_j - \kappa_i\omega), R) q_j(dy_j) = \mathcal{N}(o; \bar{T}(y_i - \mu_j - \kappa_i\omega), \tilde{R})$, and $\int e^{-\frac{\lambda}{2}\|u+s\|^2_W} \mathcal{N}(s; \bar{a}, \bar{V}) ds$ is proportional to $e^{-\frac{1}{2}\|u+\bar{a}\|^2_\Omega}$ with $\Omega = ((\lambda W)^{-1} + \bar{V})^{-1}$. The model is normalized for every value of the peer laws, which are parameters of i's model and not variables over which it infers, as in a per-agent generative model. The records with peers enter through the record term of the per-agent free energy, and the loop factors enter i's prior over its own dictionaries. Agent i's model contains no record in which i is the sender; such records belong to the receivers' models.

The recognition laws are Gaussian. In the global picture they are factorized by agent, $Q = \bigotimes_i Q_i(x_i)$, each agent's factor holding its content and the dictionaries it owns; each local update varies one factor of a single global free energy. In the per-agent picture agent i holds a Gaussian Q_i on x_i and lowers its own free energy $\mathcal{F}_i$.

Chapter 2

What conversation determines

Remark 2.1 (The interpretation gauge without linearization or Gaussian noise). Let the records be $o_e = \bar{T}_e(\bar{\theta}_e^{-1} y_i - y_j) + \varepsilon_e$ with noise of any law, independent of everything else, or more generally let each record kernel read $(y, \bar{\theta})$ only through the coded discrepancy $\bar{T}_e(\bar{\theta}_e^{-1} y_i - y_j)$. Let $K \geq 2$ and let G contain $\mathrm{GL}^+(K, \mathbb{R})$. For every configuration $(y, \bar{\theta})$ and every belief displacement $y \mapsto y'$ with $y'_i \neq 0$ and $v_e = \bar{\theta}_e^{-1} y_i + y'_j - y_j \neq 0$ on every edge $e : j \to i$, there are dictionaries $\bar{\theta}'_e \in \mathrm{GL}^+(K, \mathbb{R})$ with $\bar{\theta}'_e v_e = y'_i$, because $\mathrm{GL}^+(K, \mathbb{R})$ acts transitively on $\mathbb{R}^K \setminus \{0\}$ for $K \geq 2$. Then $\bar{\theta}'^{-1}_e y'_i - y'_j = v_e - y'_j = \bar{\theta}_e^{-1} y_i - y_j$, so $(y', \bar{\theta}')$ gives every record the same law as $(y, \bar{\theta})$. The interdependence of belief and meaning is therefore not a feature of the Gaussian tangent model: it holds pointwise, without linearization, for every record kernel of this form, and only terms other than the records can separate the two readings.

Theorem 2.2 (Belief and meaning are interdependent). In the tangent communication model the law of the comparison records depends on (y, ω) only through the coded discrepancies $L_e x = \bar{T}_e(y_i - y_j - \kappa_i \omega_e)$.

(i) The displacements $(\delta y, \delta\omega)$ that leave the law of every record unchanged form the subspace

$$\mathcal{G} = \big\{(\delta y, \delta\omega) : \bar{T}_e(\delta y_i - \delta y_j - \kappa_i \delta\omega_e) = 0 \text{ for every edge } e : j \to i\big\}. \tag{2.1}$$

If $c_i \neq 0$ for every receiver, the projection of $\mathcal{G}$ onto belief displacements is all of $(\mathbb{R}^K)^{\mathcal{V}}$: for every δy, the choice $\delta\omega_e = (\delta y_i - \delta y_j) c_i^T / |c_i|^2$ completes it to an element of $\mathcal{G}$.

(ii) The Fisher information of the records about (y, ω) vanishes exactly on $\mathcal{G}$, so along every direction of $\mathcal{G}$ the posterior precision is the sum of the contributions of the self terms, the grounded terms and the dictionary prior.

(iii) Among the displacements with $\delta\omega_e \in N_e$ for every edge, the belief components of $\mathcal{G}$ are exactly the sections $\{\delta y : \bar{T}_e(\delta y_i - \delta y_j) = 0 \text{ for every edge}\}$ of the code sheaf, whose edge condition is the displayed equality. They contain the common shifts $\delta y_i = v$.

Proof. (i) Each record is $L_e x$ plus independent noise, so its law is unchanged exactly when L_e annihilates the displacement. For the stated choice, $\kappa_i \delta\omega_e = (\delta y_i - \delta y_j) c_i^T c_i / |c_i|^2 = \delta y_i - \delta y_j$. (ii) The Fisher information of Gaussian records with fixed covariances is $\sum_e L_e^T R_e^{-1} L_e$, whose kernel is $\bigcap_e \ker L_e = \mathcal{G}$; the posterior precision is J, the sum of this form and of the other terms of eq. (1.2). (iii) For $\delta\omega_e \in N_e$, $\bar{T}_e \kappa_i \delta\omega_e = \bar{T}_e \delta\omega_e c_i = 0$, so the condition reduces to $\bar{T}_e(\delta y_i - \delta y_j) = 0$. □

An exchange that disagrees with the receiver's belief can be explained by changing the belief or by changing the dictionary. The comparison record alone does not choose between these responses. The self terms, grounded records, and anchored prior supply distinct information. Even when

grounding removes dictionary freedom modulo N_e, common content shifts remain invisible to every pairwise discrepancy.

Proposition 2.3 (Persuasion and reinterpretation). Take one edge $e : j \to i$, independent Gaussian priors $y_i \sim \mathcal{N}(m_i, \Sigma_i^0)$, $y_j \sim \mathcal{N}(m_j, \Sigma_j^0)$ and $\omega_e \sim \mathcal{N}(\bar{\omega}_e, V_e)$, and one record o_e. Put

$$r = o_e - \bar{T}_e(m_i - m_j - \kappa_i \bar{\omega}_e), \qquad C_\rho = \kappa_i V_e \kappa_i^T. \tag{2.2}$$

(i) In the global joint, with $S = \bar{T}_e(\Sigma_i^0 + \Sigma_j^0 + C_\rho)\bar{T}_e^T + R_e$, the posterior mean moves by

$$\delta y_i = \Sigma_i^0 \bar{T}_e^T S^{-1} r, \qquad \delta y_j = -\Sigma_j^0 \bar{T}_e^T S^{-1} r, \qquad \kappa_i \delta\omega_e = -C_\rho \bar{T}_e^T S^{-1} r. \tag{2.3}$$

(ii) In the law-level per-agent model of the receiver, with the sender's law $\mathcal{N}(m_j, \Sigma_j^0)$ as data, the record covariance is $\bar{T}_e(\Sigma_i^0 + C_\rho)\bar{T}_e^T + (R_e + \bar{T}_e \Sigma_j^0 \bar{T}_e^T) = S$. The receiver's mean therefore moves by $\delta y_i = \Sigma_i^0 \bar{T}_e^T S^{-1} r$ and $\kappa_i \delta\omega_e = -C_\rho \bar{T}_e^T S^{-1} r$, as in (i), and the sender does not move.

(iii) For the isotropic anchor $V_e = \sigma^2 I$, $C_\rho = \sigma^2 |c_i|^2 I$.

Proof. The quantity $u = y_i - y_j - \kappa_i \omega_e$ has prior mean $m_i - m_j - \kappa_i \bar{\omega}_e$ and covariance $\Sigma_i^0 + \Sigma_j^0 + C_\rho$, and $o_e = \bar{T}_e u + \varepsilon_e$. The covariances of y_i, y_j and $\kappa_i \omega_e$ with o_e are $\Sigma_i^0 \bar{T}_e^T$, $-\Sigma_j^0 \bar{T}_e^T$ and $-C_\rho \bar{T}_e^T$, and that of o_e is S, so Gaussian conditioning gives (i). In (ii) the term $\bar{T}_e y_j$ is distributed as $\mathcal{N}(\bar{T}_e m_j, \bar{T}_e \Sigma_j^0 \bar{T}_e^T)$ and is absorbed into the noise, which gives the displayed covariance, equal to S. For (iii), $\kappa_i^T v = v c_i^T$ and $\kappa_i (v c_i^T) = |c_i|^2 v$. □

The Gaussian conditional update assigns a discrepancy to receiver adjustment, sender adjustment, and reinterpretation through the three cross-covariances in eq. (2.3). Observation noise leaves a residual gap, so these adjustments need not reproduce the record exactly. For one record, the receiver's shifts in the law-level per-agent model coincide with the global ones; only the sender's shift differs. A plug-in receiver uses the smaller covariance $\bar{T}_e(\Sigma_i^0 + C_\rho)\bar{T}_e^T + R_e$ and hence a larger gain. The interpretation variance grows as $|c_i|^2$ under the isotropic anchor; at $c_i = 0$ the first-order dictionary term cannot absorb a discrepancy. These linear shares are features of the Gaussian family.

Corollary 2.4 (Grounding turns reinterpretation into persuasion). Let Π_e be the orthogonal projection of $\mathfrak{g}$ onto $N_e^\perp$. Then $\bar{T}_e C_\rho \bar{T}_e^T = \bar{T}_e \kappa_i \Pi_e V_e \Pi_e \kappa_i^T \bar{T}_e^T$, so only the uncertainty of the identified part of the dictionary can absorb a disagreement. If the grounded information on edge e is $g\bar{G}_e$ with $\ker \bar{G}_e = N_e$, then $\Pi_e V_e \Pi_e = O(1/g)$ and the reinterpretation share of a record is $O(1/g)$.

Proof. For $\nu \in N_e$, $\bar{T}_e \kappa_i \nu = \bar{T}_e \nu c_i = 0$, so $\bar{T}_e \kappa_i = \bar{T}_e \kappa_i \Pi_e$. With a positive-definite prior precision Λ on ω_e and grounded precision $g\bar{G}_e$, block inversion in the orthonormal bases of $N_e^\perp$ and N_e shows, by taking the Schur complement of the fixed invisible block, that the covariance restricted to $N_e^\perp$ is $O(1/g)$. □

Grounding therefore suppresses the part of the response attributable to uncertainty about the identified dictionary coordinates.

Chapter 3

The two flows

Proposition 3.1 (The global flow). Let the global recognition be $Q = \bigotimes_i \mathcal{N}(\mu_i, S_i)$ on the agent blocks $x_i = (y_i, \omega_{\mathrm{in}(i)})$, let J_{ii} be the diagonal blocks of J and $D = \bigoplus_i J_{ii}$.

(i) Up to a constant, $\mathcal{F}_G(Q) = \frac{1}{2}(\mu - x^\star)^T J(\mu - x^\star) + \frac{1}{2}\sum_i [\mathrm{tr}(J_{ii}S_i) - \log\det S_i]$, which is minimized by $S_i = J_{ii}^{-1}$ and $\mu = x^\star$.

(ii) At these covariances the Fisher-metric gradient flow of the means is $\dot{\mu} = -D^{-1}J(\mu - x^\star)$. The matrix $D^{-1}J$ is similar to the positive-definite matrix $D^{-1/2}JD^{-1/2}$, so its spectrum is real and positive, $\mathcal{F}_G$ decreases strictly away from $x^\star$, and $\mu \to x^\star$ exponentially.

(iii) Each agent's block of the force is the gradient of its local free energy, the global free energy as a function of its own factor. A record therefore moves the receiver's content, the sender's content and the receiver's dictionary through one functional.

Proof. (i) Since the factors are independent, $\mathbb{E}_Q[E] = \frac{1}{2}(\mu - x^\star)^T J(\mu - x^\star) + \frac{1}{2}\sum_i \mathrm{tr}(J_{ii}S_i) + \mathrm{const}$, and the entropy is $\frac{1}{2}\sum_i \log\det S_i + \mathrm{const}$. (ii) The Fisher metric of $\mathcal{N}(\mu_i, S_i)$ in μ_i is $S_i^{-1} = J_{ii}$, and $\nabla_\mu \mathcal{F}_G = J(\mu - x^\star)$. The similarity is $D^{-1}J = D^{-1/2}(D^{-1/2}JD^{-1/2})D^{1/2}$, and $\frac{d}{dt}\mathcal{F}_G = -\nabla\mathcal{F}_G^T D^{-1}\nabla\mathcal{F}_G$. (iii) follows by varying one factor in the same global functional; every incident likelihood contributes to that variation. □

The optimized factor means equal the Gaussian posterior mean, whereas J_{ii}^{-1} is no larger in the Loewner order than the exact marginal covariance. Indeed, the exact marginal precision is the Schur complement $J_{ii} - J_{i,-i}J_{-i,-i}^{-1}J_{-i,i} \preceq J_{ii}$. Exact factor means are a Gaussian property; the coordinate-descent identity holds more generally, without implying exact moments [1].

Proposition 3.2 (The per-agent flow). Hold the peers' covariances at a fixed point of the per-agent covariance updates, and let agent i hold $\mathcal{N}(\mu_i, S_i)$ on x_i.

(i) Up to a constant, $\mathcal{F}_i = \frac{1}{2}(\mu_i - x_i^\dagger)^T A_i(\mu_i - x_i^\dagger) + \frac{1}{2}[\mathrm{tr}(A_iS_i) - \log\det S_i]$, where $A_i = \nabla^2_{x_i} E_i$ depends on the peers' covariances but not on their means, and $A_i x_i^\dagger = b_i + \sum_{k\neq i} C_{ik}\mu_k$ is affine in the peers' means.

(ii) At $S_i = A_i^{-1}$ the Fisher flow of the means is $\dot{\mu} = -A^{-1}(H\mu - b)$ with $A = \bigoplus_i A_i$ and $H = A - C$. Its fixed points solve $H\mu = b$.

(iii) In the plug-in model, the rows of H and b that belong to dictionaries equal the corresponding rows of J and η. This equality concerns objective derivatives before Fisher preconditioning. The rows that belong to the content of agent i equal those of J and η minus the contributions $L_e^T R_e^{-1} L_e$ and $L_e^T R_e^{-1} o_e$ of the records e in which i is the sender. In the law-level model R_e is in addition replaced by $\tilde{R}_e$ and λW_γ by $\Omega_{i\gamma}$.

(iv) For an edge $e : j \to i$ the block $H_{y_i y_j}$ contains $-\bar{T}_e^T \tilde{R}_e^{-1} \bar{T}_e$, while e contributes nothing to $H_{y_j y_i}$. Hence H need not be symmetric, and its fixed point need not equal $x^\star$.

Proof. (i) and (ii) follow as in proposition 3.1, with the peers' means as data in E_i. For (iii), compare the terms of eqs. (1.2) and (1.3). A dictionary ω_e appears in the anchor, in its grounded term, in the record of e and in the loop factors, and all of these belong to the model of its owner, with the same precisions in the plug-in model; the loop factor of γ in the owner's model has the same gradient in ω_e as the single loop term of the global energy. The content y_i appears in its self term, in the records of $\mathrm{in}(i)$, which are in i's model, and in the records in which i is the sender, which are in the receivers' models only. (iv) is read off from (iii). □

Ownership removes sender-role content derivatives from the per-agent update. The equality of dictionary derivative rows holds only in the plug-in model: the law-level model uses both $\tilde{R}_e$ and $\Omega_{i\gamma}$. These derivatives are covectors. Their Fisher velocities are obtained using the block mobilities J_{ii}^{-1} and A_i^{-1}, which can differ and mix content with owned dictionary coordinates. Equal dictionary derivative rows consequently do not establish equal dictionary velocities. The dependency statement survives for general laws whenever the declared model contains only received records and the dictionary appears only in its owner's factors; equality of derivatives after marginalizing peers does not follow from ownership.

Proposition 3.3 (A directed ring). Let $K = 1$ and let $\ell \geq 3$ agents form a directed ring in which agent i receives from agent $i + 1$, indices modulo ℓ, with $\bar{T}_e = 1$, record precision $\tilde{q}$, self precision p, anchor precision $s = \sigma^{-2}$, common operating content c, and no grounded or loop term. With the own-block Gaussian covariances held at their optima, the per-agent mean flow has the eigenvalue -1 with multiplicity ℓ and the eigenvalues $-(1 - \zeta w)$ for the ℓ-th roots of unity ζ, where

$$w = \frac{\tilde{q}s}{ps + p\tilde{q}c^2 + \tilde{q}s} = \frac{\tilde{q}_{\mathrm{eff}}}{p + \tilde{q}_{\mathrm{eff}}}, \qquad \tilde{q}_{\mathrm{eff}} = \Big(\frac{1}{\tilde{q}} + \frac{c^2}{s}\Big)^{-1}. \tag{3.1}$$

The eigenvalues with $\zeta \notin \mathbb{R}$ are not real, so the per-agent flow is not a gradient flow for any Riemannian metric, whereas the global flow of the same ring has real spectrum. As $s \to \infty$, $w \to \tilde{q}/(p + \tilde{q})$; as $s \to 0$ with $c \neq 0$, $w \to 0$. If $c = 0$, then $w = \tilde{q}/(p + \tilde{q})$ for every $s > 0$.

Proof. Agent i's energy is $\frac{1}{2}p(y_i - m_i)^2 + \frac{1}{2}s\omega_i^2 + \frac{1}{2}\tilde{q}(o_i - y_i + y_{i+1} + c\omega_i)^2$. In the variables (y_i, ω_i) its Hessian is $A_i = \left(\begin{smallmatrix} p+\tilde{q} & -\tilde{q}c \\ -\tilde{q}c & s+\tilde{q}c^2 \end{smallmatrix}\right)$ with $\det A_i = ps + p\tilde{q}c^2 + \tilde{q}s$, and its coupling to (y_{i+1}, ω_{i+1}) is $C' = \left(\begin{smallmatrix} -\tilde{q} & 0 \\ \tilde{q}c & 0 \end{smallmatrix}\right)$. The flow matrix is block circulant, so on the vectors $x_k = \zeta^k v$ it acts as $-(I + \zeta A_i^{-1} C')$. The matrix $A_i^{-1}C'$ has zero second column, and its first diagonal entry is $\det A_i^{-1}[(s + \tilde{q}c^2)(-\tilde{q}) + (\tilde{q}c)^2] = -w$. Hence the eigenvalues are -1 and $-(1 - \zeta w)$. The linearization of a gradient flow at an equilibrium is self-adjoint for the metric there and has real spectrum, since its Jacobian is the negative metric inverse times the symmetric objective Hessian. The global statement is proposition 3.1(ii). The second form of w and the limits are algebra. □

For $c \neq 0$, uncertainty about the dictionary acts in the mean equations as additional record noise of variance c^2/s. The directed oscillatory modes retain angular frequency $w \operatorname{Im} \zeta$, while $0 < w < 1$ keeps every mode stable. In the limit $s \downarrow 0$ the receiver can absorb the discrepancy through reinterpretation and the coupling vanishes. At $c = 0$ the dictionary does not enter the tangent

record, so weakening its anchor does not weaken persuasion. Here $\tilde{q} = R^{-1}$ in the plug-in model and $\tilde{q} = (R + \Sigma)^{-1}$ in the law-level model at a fixed symmetric content variance Σ. The complex spectrum is one explicit instance of non-reciprocal dynamics [2]; it rules out a general gradient-flow interpretation of the per-agent picture.

Chapter 4

The grounded limit

Theorem 4.1 (Separation in the grounded limit). Let $G_e = g\bar{G}_e$ with $\bar{G}_e$ positive semidefinite and $\ker \bar{G}_e = N_e$ for every edge, keep the grounded centers $\hat{\omega}_e$ fixed, and let $g \to \infty$.

(i) In the global joint the posterior mean converges to (y^G, ω^∞), where y^G minimizes

$$E_{\text{known}}(y) = \sum_i \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_e \tfrac{1}{2}\big\|o_e + \bar{T}_e\kappa_i\hat{\omega}_e - \bar{T}_e(y_i - y_j)\big\|^2_{R_e^{-1}}, \tag{4.1}$$

the identified part of ω_e^∞ is that of $\hat{\omega}_e$, and the invisible part minimizes the dictionary prior on the coset $\hat{\omega} + \bigoplus_e N_e$. In the limit the beliefs do not depend on the dictionary prior.

(ii) In the plug-in per-agent model, if the content equations

$$P_i(y_i - m_i) - \sum_{e\in\text{in}(i)} \bar{T}_e^T R_e^{-1}\big(o_e + \bar{T}_e\kappa_i\hat{\omega}_e - \bar{T}_e(y_i - y_j)\big) = 0 \qquad (i \in \mathcal{V}) \tag{4.2}$$

have a unique solution y^P, the fixed point converges to it in the contents, and the identified dictionary parts converge to those of $\hat{\omega}_e$.

(iii) In both cases the error is $O(1/g)$.

Proof. Write $\omega_e = \hat{\omega}_e + Y_e\alpha_e + Z_e\nu_e$ in orthonormal bases Y_e of $N_e^\perp$ and Z_e of N_e. Since $\bar{T}_e\kappa_i Z_e = 0$, the records depend on (y, α) only. The energy is $\frac{g}{2}\sum_e \alpha_e^T\Gamma_e\alpha_e + \tilde{E}(y, \alpha, \nu)$ with $\Gamma_e = Y_e^T\bar{G}_eY_e$ positive definite and $\tilde{E}$ independent of g. In both pictures the stationarity equations are linear and have the block form $(g\Gamma + H_{\alpha\alpha})\alpha + H_{\alpha z}z = b_\alpha$ and $H_{z\alpha}\alpha + H_{zz}z = b_z$ with $z = (y, \nu)$ and H, b independent of g. Eliminating $\alpha = (g\Gamma + H_{\alpha\alpha})^{-1}(b_\alpha - H_{\alpha z}z) = O(1/g)$ gives $(H_{zz} + O(1/g))z = b_z + O(1/g)$, so $z \to H_{zz}^{-1}b_z$ at rate $1/g$ whenever H_{zz} is invertible. At $\alpha = 0$ the records involve y alone and the dictionary prior involves ν alone, so H_{zz} is block diagonal. Its content block gives eq. (4.1) in the global joint, where J is positive definite, and eq. (4.2) in the plug-in model, invertible by hypothesis. Its invisible-dictionary block is the restriction of the dictionary prior precision $\sigma^{-2}I + \lambda B^TWB$ to the invisible directions, positive definite, in both pictures by proposition 3.2(iii). □

The theorem identifies two different known-dictionary limits. The global mean minimizes a symmetric quadratic comparison energy, while the plug-in per-agent fixed point solves receiver-only equations. The identified dictionaries agree in the limit, but the contents need not. The theorem makes no claim about law-level covariance updates that vary with g; controlling those updates is an additional obligation.

Chapter 5

Confidence under ownership

Mean agreement does not imply agreement about uncertainty. The following example isolates one coordinate that records and grounding do not read, so all information coupling it to other dictionaries comes from one declared loop factor.

Proposition 5.1 (Echo around a loop). Let $\lambda > 0$ and let $\ell \geq 2$ edges form one loop, with distinct receivers and one coordinate z_e per edge that no record or grounded term reads, with anchor $\mathcal{N}(0, \sigma^2)$ and loop factor $\exp(-\frac{\lambda}{2}(\sum_e z_e)^2)$. Every recognition below has mean zero, and the variances of z_e are as follows.

(i) Exact joint posterior: $\sigma^2(1 + (\ell - 1)\lambda\sigma^2)/(1 + \ell\lambda\sigma^2)$, which tends to $\sigma^2(\ell - 1)/\ell$ as $\lambda \to \infty$.

(ii) Receiver-factorized global recognition: $(\sigma^{-2} + \lambda)^{-1}$, which tends to zero.

(iii) Per-agent law-level recognition at its symmetric covariance fixed point: the unique positive solution v of

$$v^{-1} = \sigma^{-2} + \big(\lambda^{-1} + (\ell - 1)v\big)^{-1}, \tag{5.1}$$

which tends to $\sigma^2(\ell - 2)/(\ell - 1)$ for $\ell \geq 3$ and satisfies $v = \sigma\lambda^{-1/2}(1 + o(1))$ for $\ell = 2$.

(iv) Cavity recognition, in which each receiver's loop source uses its partners' laws with its own contribution removed, so that those laws are the anchors: $(\sigma^{-2} + (\lambda^{-1} + (\ell - 1)\sigma^2)^{-1})^{-1}$, which equals (i) for every λ.

Proof. All terms are even in z, so the means vanish. (i) The precision is $\sigma^{-2}I + \lambda\mathbf{1}\mathbf{1}^T$, and the Sherman–Morrison formula gives the diagonal entry $\sigma^2 - \sigma^4\lambda/(1 + \ell\lambda\sigma^2)$. (ii) The factorized covariance is the inverse of the diagonal of the precision. (iii) By symmetry each receiver's loop source is the convolution of the loop factor with $\ell - 1$ partner laws of variance v, which gives eq. (5.1). With $\epsilon = \lambda^{-1}$, clearing denominators gives $(\ell - 1)\sigma^{-2}v^2 + (\epsilon\sigma^{-2} + 2 - \ell)v - \epsilon = 0$, whose roots have product $-\epsilon\sigma^2/(\ell - 1) < 0$, so exactly one is positive. As $\epsilon \to 0$ it tends to $\sigma^2(\ell - 2)/(\ell - 1)$ for $\ell \geq 3$, and for $\ell = 2$ it satisfies $v^2 + \epsilon v - \epsilon\sigma^2 = 0$, so $v = \sigma\epsilon^{1/2}(1 + o(1))$. (iv) With $\epsilon = \lambda^{-1}$, $(\sigma^{-2} + (\epsilon + (\ell - 1)\sigma^2)^{-1})^{-1} = \sigma^2(\epsilon + (\ell - 1)\sigma^2)/(\epsilon + \ell\sigma^2)$, which is (i) after multiplying numerator and denominator by λ. □

All four constructions have the same zero mean and different variances. Factorized reverse-KL recognition suppresses the correlations by which the exact law satisfies the sum constraint. The per-agent update instead treats a partner's current marginal as new independent information, even though that marginal already contains information received from the same loop. For a two-cycle its fixed-point variance tends to zero while the exact marginal tends to $\sigma^2/2$. Removing the loop contribution from the peer messages gives the anchor laws and restores the exact marginal here. This is belief propagation on the tree consisting of one factor connected to ℓ variables [3]; it is not a general exactness theorem for overlapping loop factors.

The recycled-information mechanism can occur for non-Gaussian laws as well. The displayed fixed points, rates, and equality of factor means are conclusions only for this Gaussian single-loop model.

Chapter 6

The reciprocal potential

The reciprocal potential is a function of agent laws. It moves each source through the costs of receivers that attend to it. The following extension includes dictionary laws as configuration variables. Registered comparison records are absent from this potential; adding persistent likelihood terms would define another objective.

Definition 6.1 (Tangent reciprocal potential). Let a configuration consist of content laws $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and a factorized dictionary law $r = \bigotimes_e \mathcal{N}(a_e, V_e)$. For an edge $e : j \to i$ put $S_{i,e} = \bar{T}_e \Sigma_i \bar{T}_e^T + R_e$ and $S_{j,e} = \bar{T}_e \Sigma_j \bar{T}_e^T + R_e$, and define the message laws

$$m_e^{\mathrm{rec}}(\omega_e) = \mathcal{N}\big(\bar{T}_e(\mu_i - \kappa_i \omega_e), S_{i,e}\big), \qquad m_e^{\mathrm{src}} = \mathcal{N}\big(\bar{T}_e \mu_j, S_{j,e}\big), \tag{6.1}$$

the law of the message the receiver would send about its own content read in the sender's coordinates, and the law of the sender's message. With fixed positive rows β_e, the tangent reciprocal potential is

$$\mathcal{E} = \sum_i D_{\mathrm{KL}}\big(q_i \| \mathcal{N}(m_i, P_i^{-1})\big) + \sum_e \beta_e \mathbb{E}_r\big[D_{\mathrm{KL}}\big(m_e^{\mathrm{rec}}(\omega_e) \| m_e^{\mathrm{src}}\big)\big] + D_{\mathrm{KL}}(r \| \pi_\omega), \tag{6.2}$$

where π_ω is the normalized dictionary prior with density proportional to $\exp(-\frac{1}{2}\omega^T(\sigma^{-2} I + \lambda B^T W B)\omega - \frac{1}{2}\sum_e \|\omega_e - \hat{\omega}_e\|_{G_e}^2)$.

The receiver's law comes first and the source's second, as in a directed peer divergence, and the costs are divergences between message laws, rather than comparisons between independent content samples. The self term replaces the local term $\mathcal{E}_0$ by its tangent form. The last term makes the dictionary laws configuration variables whose part of the potential is a recognition free energy over dictionaries. For normalized variable rows with priors π_i and temperatures T_i, use $\mathcal{E}_T = \mathcal{E} + \sum_i T_i D_{\mathrm{KL}}(\beta_i \| \pi_i)$. Its Gibbs reduction E_T has the derivative of the fixed-row potential with the rows evaluated at their Gibbs values, by the envelope identity, so the derivative statements below apply pointwise with those weights inserted. Substituting the Gibbs rows into the entropy-free weighted sum does not give this derivative. State-dependent Gibbs weights do not preserve the constant-coefficient mean-flow theorem.

Proposition 6.2 (Forces in the reciprocal potential). Hold the covariances Σ_i and V_e fixed. Let E^S be the global energy eq. (1.2) with every R_e replaced by $S_{j,e}/\beta_e$ and every registered value o_e by zero, with precision J^S and information vector η^S.

(i) $\mathcal{E} = E^S(\mu, a) + c(\Sigma, V)$, where c does not depend on the means.

(ii) The force on a dictionary mean a_e comes only from the cost of its own edge and from π_ω, because no other cost contains ω_e. There is no additional sender-role edge derivative for the

dictionary. The force on a content mean contains the terms of the edges on which the agent is the sender, as in the global joint.

(iii) The Fisher gradient flow of the means is $(\dot{\mu}, \dot{a}) = -MJ^S\big((\mu, a) - x^S\big)$ with $M = (\bigoplus_i \Sigma_i) \oplus (\bigoplus_e V_e)$ and $x^S = (J^S)^{-1}\eta^S$. Its spectrum is real and negative, and its stationary point is the posterior mean of the global joint in which every record has noise $S_{j,e}/\beta_e$ and registered value zero.

(iv) For $\beta_e = 1$, the content rows of J^S minus the sender terms are the content rows of the per-agent law-level model with $o_e = 0$ at the same covariances. The dictionary rows of J^S differ from the per-agent law-level ones only in that the loop sources of the latter are widened as in eq. (1.4).

Proof. (i) For Gaussian laws $D_{\mathrm{KL}}(\mathcal{N}(\alpha, A)\|\mathcal{N}(\beta', B)) = \frac{1}{2}[\mathrm{tr}(B^{-1}A) - n + (\alpha - \beta')^T B^{-1}(\alpha - \beta') + \log\det B - \log\det A]$. Under r the mean of m_e^{rec} is $\bar{T}_e(\mu_i - \kappa_i a_e) - \bar{T}_e\kappa_i(\omega_e - a_e)$, so the averaged cost is

$$\tfrac{1}{2}\big[\mathrm{tr}(S_{j,e}^{-1}S_{i,e}) - r_e + d_e^T S_{j,e}^{-1} d_e + \mathrm{tr}(S_{j,e}^{-1}\bar{T}_e\kappa_i V_e \kappa_i^T \bar{T}_e^T) + \log\det S_{j,e} - \log\det S_{i,e}\big] \tag{6.3}$$

with $d_e = \bar{T}_e(\mu_i - \mu_j - \kappa_i a_e) = L_e(\mu, a)$. The self term is $\frac{1}{2}(\mu_i - m_i)^T P_i(\mu_i - m_i)$ plus covariance terms. Since r is a product and $\mathbb{E}_r[\omega^T A\omega] = a^T Aa + \mathrm{tr}(AV)$, $D_{\mathrm{KL}}(r\|\pi_\omega)$ is $\frac{1}{2}a^T(\sigma^{-2}I + \lambda B^T W B)a + \frac{1}{2}\sum_e (a_e - \hat{\omega}_e)^T G_e (a_e - \hat{\omega}_e)$ plus covariance terms and a constant. The mean-dependent terms sum to E^S. (ii) In eq. (6.2) the variable ω_e appears only in the cost of e and in π_ω. (iii) The Fisher metric of q_i in μ_i is Σ_i^{-1} and that of $\mathcal{N}(a_e, V_e)$ in a_e is V_e^{-1}, so the Fisher gradient is $M\nabla E^S = MJ^S((\mu, a) - x^S)$, and MJ^S is similar to $M^{1/2}J^S M^{1/2}$, which is positive definite because the self terms and the anchor are. (iv) In the per-agent law-level model $\tilde{R}_e = R_e + \bar{T}_e\Sigma_j\bar{T}_e^T = S_{j,e}$; the rest is proposition 3.2(iii). □

At fixed covariances and fixed rows, the reciprocal potential has the same mean-dependent quadratic as a particular widened Gaussian global model. This identity does not identify their covariance objectives, their recognition families, or their full dynamics. A dictionary occurs only in its own edge cost and in the declared prior, so other dictionary means enter its derivative only through the loop term. This locality is a dependency property of the declared fixed codes and prior; it persists for other laws with the same factor structure. The explicit quadratic and spectral conclusions use Gaussian laws and fixed covariances.

Proposition 6.3 (Disagreement and certainty).

(i) In the Gaussian tangent forms of the global joint and of both per-agent models, the optimizing recognition covariances do not depend on the registered records or on any mean. This is a feature of linear-Gaussian models, whose posterior covariance never depends on the data, and not a property of the free energies.

(ii) In the tangent reciprocal potential, the derivative of the unweighted cost of an edge $e : j \to k$ with respect to the sender's covariance Σ_j is

$$\tfrac{1}{2}\bar{T}_e^T S_{j,e}^{-1}\big[S_{j,e} - S_{k,e} - d_e d_e^T - \bar{T}_e\kappa_k V_e \kappa_k^T \bar{T}_e^T\big] S_{j,e}^{-1}\bar{T}_e, \tag{6.4}$$

so the source-role force on the sender's covariance vanishes exactly when the sender's message covariance equals the receiver's predicted message covariance plus the outer product of the coded disagreement and the reinterpretation covariance.

(iii) Let $K = 1$, $\bar{T}_e = 1$, $\beta_e = 1$, and hold the dictionary variance $V_e \geq 0$ fixed. Hold the means, $R > 0$, and receiver variance v_k fixed, and let the sender's variance-dependent objective consist only of its self term with precision p_j and one outgoing edge cost. With $a = v_k + R + d^2 + c_k^2 V_e$, the stationary sender variance v solves $f(v) = p_j - v^{-1} + (v + R)^{-1} - a(v + R)^{-2} = 0$. This equation has exactly one positive root, which is the unique global minimizer of the variance-dependent objective, and it increases strictly with $d^2 + c_k^2 V_e$.

Proof. (i) The global covariances are J_{ii}^{-1} and the per-agent ones are A_i^{-1}; these matrices depend on the precisions and on the peers' covariances only. (ii) The terms of the averaged cost that contain $S_{j,e}$ are $\frac{1}{2}[\mathrm{tr}(S_{j,e}^{-1} C) + \log\det S_{j,e}]$ with $C = S_{k,e} + d_e d_e^T + \bar{T}_e \kappa_k V_e \kappa_k^T \bar{T}_e^T$; their derivative with respect to $S_{j,e}$ is $\frac{1}{2}(S_{j,e}^{-1} - S_{j,e}^{-1} C S_{j,e}^{-1})$, and $S_{j,e} = \bar{T}_e \Sigma_j \bar{T}_e^T + R_e$ gives eq. (6.4) by the chain rule. (iii) The function f is twice the derivative of $\mathcal{E}$ in v. Over the common denominator $v^2(v + R)^3$ its derivative in v has numerator $R^3 + 3R^2 v + 4Rv^2 + 2(a - R)v^2 > 0$. As $f \to -\infty$ when $v \downarrow 0$ and $f \to p_j > 0$ as $v \to \infty$, f has exactly one root. Its derivative with respect to a is $-(v + R)^{-2} < 0$, so the implicit function theorem gives $dv/da > 0$. □

In the potential the source law is compared only through its message law. For a source q_η in a regular exponential family with statistic t, compared through a fixed kernel $\mathcal{K}$ with positive message densities, $\partial_\eta D_{\mathrm{KL}}(p \| q_\eta \mathcal{K}) = \mathbb{E}_{q_\eta} t - \mathbb{E}_{\tilde{p}_\eta} t$ with $\tilde{p}_\eta = \int p(dh) q_\eta(\cdot \mid h)$, whenever the moments exist and differentiation under the integral is justified. Source-role stationarity is therefore the fixed-point condition of the M-step of the EM algorithm: the source matches, in its sufficient statistics, its own posterior reconstruction of the receiver's message law. For an identity kernel it is the moment match $\mathbb{E}_{q_\eta} t = \mathbb{E}_p t$. The Gaussian covariance formula is the second-moment instance at fixed mean, including the squared discrepancy about the sender's mean. Its contribution is multiplied by β_e in the full potential. The scalar result proves increasing stationary variance with disagreement only while the other specified quantities are frozen. Competing self terms, receiver roles, other source roles, moving means, and changing rows determine the full covariance velocity, so no monotone loss of certainty is asserted for that flow. General non-Gaussian Bayesian posteriors may also change their uncertainty when records change.

For product dictionary laws in definition 6.1, covariance stationarity gives $V_e^{-1} = \sigma^{-2} I + \lambda(B^T W B)_{ee} + G_e + \beta_e \kappa_i^T \bar{T}_e^T S_{j,e}^{-1} \bar{T}_e \kappa_i$. The grounded and record information vanish on N_e. When $W_\gamma = I$, the loop diagonal block is the number of simple loops containing e times the identity, so N_e is invariant under this precision. In the single-loop setting of proposition 5.1, the variance on an unread coordinate is $(\sigma^{-2} + \lambda)^{-1}$, the same value as factorized global recognition. No general marginal-variance formula for overlapping loops is implied.

Chapter 7

Scope and open problems

Open problem 7.1 (Rows and covariances in the reciprocal potential). The main manuscript chapter gives the complete Gaussian tangent flow of means, covariances and rows, its global forward existence and convergence, and, for one three-agent example and its reference data, the local critical mode as a subcritical pitchfork. Determine its distant stationary branches and long-time destinations, the corresponding behavior on graphs with cycles, and well-posedness beyond regular finite-dimensional law families. Neither the scalar certainty result nor that local instability establishes grouping or formation.

Open problem 7.2 (Convergence of per-agent flows). Find conditions on the self, record, and loop precisions under which the per-agent fixed point is unique and its mean flow converges. The law-level covariance updates have a unique attracting fixed point, by a monotonicity and subhomogeneity argument given in the main manuscript chapter; the open part concerns the means, whose plug-in flow can be unstable. The directed ring supplies one stable non-gradient example, not a general convergence theorem.

Open problem 7.3 (Beyond the tangent model). Keep the product $\omega_e(y_i - c_i)$ and let the operating contents follow the beliefs. Factorized recognition then acquires second-moment terms, and the interpretation gauge of theorem 2.2 becomes a nonlinear orbit. Determine whether grounding still separates belief from meaning at the nonlinear level.

Open problem 7.4 (General laws). The Gaussian tangent model is a computational simplification. Extend its conclusions to general laws, with record kernels that read the coded discrepancy and location-family noise. Four parts are open: the attribution of a coded disagreement among receiver persuasion, sender persuasion and reinterpretation when contents and dictionary priors are not Gaussian; the grounded limit, in which the reinterpretation share should vanish as grounding identifies the dictionaries; whether disagreement changes certainty in the two free energies, and how that Bayesian effect compares with the source-role moment matching of the reciprocal potential; and the echo and cavity statements for non-Gaussian loop factors, where cavity messages are belief propagation without Gaussian closure and the factorized means are no longer exact.

Open problem 7.5 (What agents must exchange). A receiver using a loop prior needs information about dictionaries owned elsewhere. An echo-free protocol must specify which prior contributions are removed from each transmitted law. Formulate that exchange through declared codes and record kernels, and determine the loop information it makes observable.

Bibliography

[1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006.

[2] Michel Fruchart, Ryo Hanai, Peter B. Littlewood, and Vincenzo Vitelli. Non-reciprocal phase transitions. *Nature*, 592:363–369, 2021.

[3] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, San Mateo, CA, 1988.